

1.13 Vectors

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs v1–v7

What this topic covers. Vectors in two and three dimensions — notation, addition and scalar multiplication, magnitude and direction, position vectors and distances — and their use in geometric problems and in resolving forces.

Not in this specification: the scalar (dot) product, and the vector equation of a line. Both appear in other A-Level Mathematics specifications but are *not* part of H640, so questions here will not require them. Angles between vectors are handled by trigonometry, not by a dot product.

1. Notation and language

Term	Meaning
Scalar	A quantity with magnitude only — mass, time, speed.
Vector	A quantity with magnitude <i>and</i> direction — displacement, velocity, force.
Modulus / magnitude	The size of a vector, written $ \mathbf{a} $ or a .
Unit vector	A vector of magnitude 1.
Position vector	The vector from the origin to a point.
Equal vectors	Same magnitude and same direction — position is irrelevant.
Parallel vectors	One is a scalar multiple of the other.

Collinear points

Points lying on the same straight line.

Vectors are printed in **bold** and handwritten underlined. In component form, in two dimensions:

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

and in three dimensions the unit vector **k** is added along the *z*-axis (spec ref v7):

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$$

2. Arithmetic with vectors

Addition and subtraction work component by component, and a scalar multiple scales the magnitude while preserving direction (reversing it if the scalar is negative).

Worked example. With $\mathbf{a} = 4\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = \mathbf{i} - 2\mathbf{j}$:

$$\mathbf{a} + \mathbf{b} = 5\mathbf{i} - \mathbf{j}, \quad \mathbf{a} - \mathbf{b} = 3\mathbf{i} + 3\mathbf{j}, \quad 3\mathbf{a} = 12\mathbf{i} + 3\mathbf{j}$$

Geometrically, addition is the triangle (or parallelogram) rule: place the vectors nose to tail, and the resultant runs from the start of the first to the end of the last.

3. Magnitude and direction

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2} \quad (2D), \quad |\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad (3D)$$

Worked example. $|3\mathbf{i} + 4\mathbf{j}| = \sqrt{9 + 16} = 5$; $|2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}| = \sqrt{4 + 9 + 36} = 7$.

Unit vectors

To find a unit vector in the direction of **a**, divide by its magnitude:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Worked example. A unit vector in the direction of $6\mathbf{i} - 8\mathbf{j}$: the magnitude is 10, so $\hat{\mathbf{a}} = 0.6\mathbf{i} - 0.8\mathbf{j}$.

Magnitude-direction form

In two dimensions a vector may be given by its magnitude r and the angle θ it makes with the positive x -axis (v3):

$$a_1 = r \cos \theta, \quad a_2 = r \sin \theta, \quad r = \sqrt{a_1^2 + a_2^2}, \quad \tan \theta = \frac{a_2}{a_1}$$

Worked example. Express $3\mathbf{i} + 3\mathbf{j}$ in magnitude-direction form.

$r = \sqrt{18} = 3\sqrt{2}$ and $\tan \theta = 1$, so $\theta = 45^\circ$.

`arctan` always returns an angle in the first or fourth quadrant. For a vector such as $(-3, -4)$, which points into the **third** quadrant, the calculator gives 53.1° and you must add 180° to get 233.1° . Always sketch the vector before quoting a direction.

4. Position vectors and distance

If A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$$

and the distance between them is $|\mathbf{b} - \mathbf{a}|$ (v4, v5).

Worked example. $A(2, 1)$ and $B(6, 4)$. Then $\overrightarrow{AB} = 4\mathbf{i} + 3\mathbf{j}$ and $|AB| = \sqrt{16 + 9} = 5$.

Worked example. $A(2, -1, 4)$ and $B(5, 3, 0)$. Then $\overrightarrow{AB} = 3\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$, so $|AB| = \sqrt{9 + 16 + 16} = \sqrt{41}$.

The order matters: $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, "destination minus start". Reversing it gives a vector of the same magnitude in the opposite direction — harmless for a distance, fatal for a direction.

Midpoints and division of a line

The midpoint of AB has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{b})$. More generally, the point dividing AB in the ratio $m : n$ has position vector

$$\frac{n\mathbf{a} + m\mathbf{b}}{m + n}$$

Worked example. P divides AB in the ratio $2 : 1$. Then $\overrightarrow{OP} = \mathbf{a} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a} + 2\mathbf{b}}{3}$.

5. Geometric problems

Proving points collinear

Worked example. A, B, C have position vectors $\mathbf{i} + \mathbf{j}$, $3\mathbf{i} + 5\mathbf{j}$, $5\mathbf{i} + 9\mathbf{j}$. Show they are collinear.

$$\overrightarrow{AB} = 2\mathbf{i} + 4\mathbf{j}, \quad \overrightarrow{BC} = 2\mathbf{i} + 4\mathbf{j}$$

$\overrightarrow{BC}$ is a scalar multiple of $\overrightarrow{AB}$ (here equal to it), so the two are parallel; since they share the point B , the three points are collinear.

Parallel alone is not enough — two parallel vectors can lie on different lines. You must also state the **common point**. That sentence is what completes the proof.

Parallel vectors in three dimensions

Worked example. Find λ so that $2\mathbf{i} + 4\mathbf{j} + \lambda\mathbf{k}$ is parallel to $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

The first must be a scalar multiple of the second. Comparing $\mathbf{i}$ components gives the factor 2, so $\lambda = 2 \times 3 = 6$.

Shapes

In a parallelogram $OACB$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, the diagonal is $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$. The centroid of a triangle with vertices $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ has position vector $\frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$.

6. Vectors and forces

The specification explicitly includes using vectors in context, "including problems involving forces", and interpreting the sum of force vectors as the **resultant** force (v6). This is the bridge into Mechanics.

Worked example. Forces $3\mathbf{i} + 2\mathbf{j}$ and $-\mathbf{i} + 4\mathbf{j}$ act on a particle. Find the resultant.

$$(3 - 1)\mathbf{i} + (2 + 4)\mathbf{j} = 2\mathbf{i} + 6\mathbf{j}$$

Worked example. Forces $\mathbf{F}_1 = 2\mathbf{i} + 3\mathbf{j}$, $\mathbf{F}_2 = -5\mathbf{i} + \mathbf{j}$ and $\mathbf{F}_3$ hold a particle in equilibrium. Find $\mathbf{F}_3$.

Equilibrium means the vector sum is zero, so

$$\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2) = -(-3\mathbf{i} + 4\mathbf{j}) = 3\mathbf{i} - 4\mathbf{j}$$

7. A useful inequality

For any two vectors, the magnitude of the sum depends on their directions:

$$||\mathbf{a}| - |\mathbf{b}|| \leq |\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$$

So if $|\mathbf{a}| = 5$ and $|\mathbf{b}| = 12$, then $|\mathbf{a} + \mathbf{b}|$ lies between 7 and 17 — reaching 17 only when they point the same way, and 7 only when they oppose.

Magnitudes do **not** simply add: $|\mathbf{a} + \mathbf{b}| \neq |\mathbf{a}| + |\mathbf{b}|$ in general. Adding the magnitudes is one of the most frequent vector errors.

8. Common pitfalls

- Writing $\overrightarrow{AB} = \mathbf{a} - \mathbf{b}$ instead of $\mathbf{b} - \mathbf{a}$.
- Adding magnitudes instead of adding vectors and then taking the magnitude.
- Quoting an $\arctan$ angle without checking the quadrant.
- Proving two vectors parallel and stopping, without stating the common point, in a collinearity proof.
- Forgetting the third component when computing a magnitude in three dimensions.
- Treating a scalar multiple as changing direction when the scalar is positive — it does not.
- Reaching for a dot product, which is not part of this specification.